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The efficiency effects of bank mergers: An overview of case studies of nine mergers

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Abstract

This paper summarizes nine case studies, by nine authors, on the efficiency effects of bank mergers. The mergers selected for study were ones that seemed relatively likely to yield efficiency gains. That is, they involved relatively large banks generally with substantial market overlap, and most occurred during the early 1990s when efficiency was getting a lot of attention in banking. All nine of the mergers resulted in significant cost cutting in line with premerger projections. Four of the nine mergers were clearly successful in improving cost efficiency but five were not. It is not possible to isolate specific factors from these mergers that are most likely to yield efficiency gains, but the most frequent and serious problem was unexpected difficulty in integrating data processing systems and operations. © 1998 Elsevier Science B.V. All rights reserved.

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1. Overview

The possibility that bank mergers, especially horizontal (in-market) mergers, will yield efficiency gains is an issue that has been subject to considerable debate. This paper summarizes nine case studies that address this issue and describes the motivation for, methodology, and findings of the studies.²

Since the mid-to-late 1980s, many bankers and bank analysts have argued that bank mergers result in efficiency gains (Krabill, 1985; Meehan, 1989; McNamee, 1992). On the other side, recently some analysts have expressed skepticism.³ Furthermore, the bulk of empirical research shows no evidence of efficiency gains from bank mergers⁴ or from increased bank size per se (that is, due to scale efficiencies) beyond a small size.⁵ Even those few studies that have analyzed the efficiency effects of horizontal mergers, the type of merger thought to be the most likely to yield efficiency gains, have found that such mergers do not, on average, yield efficiency gains (Azarchs, 1995; Srinivasan and Wall, 1992; Berger and Humphrey, 1992; Rhoades, 1993); nor have “event” studies of bank mergers.⁶ Such conclusions from the typical cross-section studies are generalizations based on statistical tests, but there are exceptions in which mergers do result in efficiency gains. Nevertheless, the marketplace appears to have generally questioned the gains from in-market bank mergers as reflected in the adverse effect of a merger announcement on acquirers’ stock prices.⁷

² The authors of the studies are Dean Amel, Jim Burke, Anthony Cynrak, Timothy Hannan, Robert Kurtz, Nellie Liang, Steve Pilloff, Robin Prager, and Donald Savage. Frederick Schroeder made significant contributions to two of the case studies and Jim Berkovec to one. Anthony Cynrak also contributed to developing the ratios and data, and reviewing the studies. The studies average about 50 pages in length. In order to maintain confidentiality of certain data, and because it was deemed inappropriate to publish evaluations and comparisons of specific institutions by Federal Reserve Board staff, the individual studies and the identities of the institutions are not being made available.

³ For example, one cautioned that in spite of pressures from brokerage firms on banks to cut costs in order to show positive short-term results, a single-minded focus on cost cutting may be counterproductive in the longer run with respect to providing customer service, engaging in certain product lines, and collecting core deposits (see Asher, 1994; Marks, 1991; Klinkerman, 1991; Matthews, 1993).

⁴ All 39 of the studies of bank mergers and performance published between 1980 and 1993 are summarized in Rhoades (1994).

⁵ A good review of the rather large literature on scale economies may be found in Humphrey (1990).

⁶ See, for example, Hannan and Wolken (1989). Other event studies in banking are cited and briefly summarized in Rhoades (1994).

⁷ See SNL Securities (1993). The neutral or adverse effect of a merger announcement on acquirers’ stock prices has been a fairly general result for mergers of industrial firms as well as banks (see a special issue on the market for corporate control, and especially the lead article by Jensen and Ruback, 1983).

Because of the continuing disagreement between most systematic empirical studies and the views of some bankers, and because of the relevance of the issue for individual bank strategy, industry performance, and public policy, the nine studies revisited the issue. They used a case study methodology rather than the cross-section statistical methodology used in most earlier studies.

The nine mergers studied were not randomly selected. Indeed, the mergers selected were generally large horizontal mergers that are thought to be the kind of merger most likely to yield efficiency gains. These deals occurred since the mid-to-late 1980s, most during the early 1990s, during which time there had been considerable emphasis in the industry on cutting costs. Due to the non-random selection of bank mergers for analysis, it will not be possible to generalize from the findings, to all mergers. Nevertheless, the findings may point toward what conditions surrounding a merger are most likely to yield efficiency gains and why efficiency gains are, or are not, realized.

2. Methodology

In studying the issue of possible gains from mergers, it is important to distinguish between cost reductions and efficiency improvements; they are not synonymous. Reductions in operating expenses may result from cutting employees, closing branches, consolidating headquarters offices, closing computer and back-office operations, and so forth. Such reductions in expenses, however, do not automatically translate into improvements in efficiency as measured by an expense ratio, such as expenses to assets or revenues. Reductions in expenses may be accompanied by corresponding reductions in assets and revenues, which simply represent shrinkage of the firm rather than efficiency improvements.⁸ An improvement in efficiency requires that costs be reduced by more than any decline in assets (revenues). Failure to distinguish between cost reductions and efficiency gains may at least partly explain the difference in views between bankers, who often emphasize the *cost* reductions to be achieved from mergers, and researchers, who generally study the *efficiency* effects of mergers. The individual case studies and this summary analyze the *efficiency* effects of mergers because of the economic significance of efficiency and its amenability to interpretation and analysis.

⁸ This phenomenon was suggested by Linder and Crane (1992).

2.1. *Case study approach*

Most earlier research on the efficiency effects of mergers used a cross-section analysis.⁹ That type of analysis typically includes a relatively larger number of mergers and the use of a statistical model. The great appeal of the cross-section approach is that it permits statistical tests that control for various other influences on merger performance and, as a result, statistically valid generalizations may be made. Despite the virtues of the cross-section methodology, criticisms of this methodology for not adequately capturing industry-specific or firm-specific idiosyncrasies have resulted in the re-emergence of the analysis of particular industries or firms in industrial organization.¹⁰

Because of the limited number of observations (often only one industry or firm) case studies do not permit statistically valid generalizations. However, the case study methodology may provide insights into firm (industry) behavior and performance that cannot be captured in a cross-section study because a case study may use a wide range of data and institutional detail from sources that may be unique to a firm, or industry. Such information may allow explanations for observed behavior and performance and help to identify situations to which the cross-section generalizations do not apply. In this way, the case studies in this volume may help resolve the continuing debate over the merger efficiency issue.

2.2. *Planning meetings*

In the initial stages of this project, the economists conducting the studies met with bank consultants and stock analysts from eight different firms to get ideas for designing the project.¹¹ The interviews were conducted during the fall 1991 and typically lasted 1–2 hours. There was a substantial degree of consistency in the views of the interviewees. Particularly notable observations were as follows.

1. Roughly one-half of savings from mergers will occur during the first year, and all savings will be achieved within three years.

⁹ A notable exception is Crane and Linder (1993). The case study approach was also used in assessing the performance of four merger-active bank holding companies. These four case studies focus on merger-active companies rather than specific mergers to assess the effects of mergers (see Johnson, 1993). Another set of case studies of bank mergers focuses more on the strategies and motivations than on the efficiency effects (see Calomaris and Karceski, 1995).

¹⁰ Many of these use a time series analysis or some other statistical procedure, which distinguishes them from traditional case studies. This may imbue them with some of the strengths and some of the weaknesses of both types of studies. This trend is illustrated by a special issue of the *Journal of Industrial Economics* (1987) and Geroski (1988). Also see Tirole (1988), Schmalensee (1989), and Bresnahan (1989).

¹¹ The interviewees were very helpful, and their time and expertise is appreciated. They are Danielson Associates, Furash and Co., Keefe Bruyette, The MAC Group, McKinsey and Co., Montgomery Securities, Oliver–Wyman Associates, and Secura.

2. Most significant cost savings could be accomplished without merger.
3. Any cost saving or efficiency gain should be observable in public financial data such as the Call Reports.

2.3. Selection of merger cases

The mergers analyzed were not selected randomly. Indeed, the mergers were generally selected on the basis of characteristics that, according to some industry observers, should cause a merger to result in efficiency gains. Those characteristics are that the merger involved: (1) large firms, (2) firms with considerable office overlap, and (3) mergers that occurred in recent years, during which time cost cutting and efficiency received a great deal of attention in banking as well as other industries.¹²

Based on these general characteristics, several specific criteria were applied in selecting mergers for analysis. First, the acquiring firm was generally required to have at least \$10 billion in assets at the time of merger, and the target was generally required to be at least half as large as the acquiring firm.¹³ The requirement that the target be fairly large in relation to the acquirer is an attempt to ensure that the target would be of sufficient size that any efficiency or performance effect of its being acquired would be reflected in data for the combined firm. Second, the merging parties were generally required to have significant office overlap. That is, both were required to have offices in at least one market that is important to their operations. No specific numerical criteria were applied with respect to the office overlap requirement. The office overlap criterion was employed because closing “redundant”, or directly competing, offices is believed to be a source of cost reductions. Third, seven of the nine mergers selected had taken place since 1990, by which time there was much emphasis on cost cutting in the industry.

3. Study framework and data

The same basic analytical framework was employed in all of the case studies. For example, all analyzed, at a minimum, a common set of financial ratios, three econometric cost measures, and the effect of the merger announcement on the stock price of the acquiring and acquired firms. A general conformity

¹² These characteristics were cited variously in our initial interviews with consultants and analysts, and in the press (see, for example, Meehan, 1989, 1990; Meehan et al., 1991; McNamee et al., 1991; McNamee, 1992; Bremner and Zachary, 1991).

¹³ This criterion was not met in one of the mergers in that the acquirer was several times larger than the target.

in the approach of these case studies was adopted to ensure that any differences in findings among the studies reflect different merger results rather than differences in analytical approach. Despite the conformity in approach adopted for the studies, investigators were encouraged to be innovative in order to address unique characteristics of the mergers being studied.

All of the studies analyze at least 16 financial ratios, including seven expense ratios, two profitability ratios, and five balance sheet ratios. The expense and profitability ratios were used to analyze efficiency and profitability during the pre- and post-merger periods. The balance sheet ratios provided information on other changes that may have occurred, aside from or as a result of the merger, that might have affected efficiency or profitability.

All ratios were analyzed for three years preceding the year of the merger and three years after the merger. The three-year time period was used because of the almost unanimous agreement among the experts we interviewed that about half of any efficiency gains should be apparent after one year, and all gains should be realized within three years.

Several sets of ratios were examined in detail for each of the three years before and after merger. For the pre-merger period, ratios for both the acquirer and the target were examined to get an indication as to the relative efficiency and performance of the acquirer and target. This may be of special interest, because common sense suggests, and several of the industry experts we interviewed stated, that a merger is most likely to result in efficiency gains if the acquirer is more efficient than the target.¹⁴ In addition, for the pre-merger period, a hypothetical combined firm was created by calculating simple average ratios from the data for the acquirer and target. Finally, for the pre-merger period, ratios for a control, or peer, group of other firms were examined. These ratios, generally constructed as a simple average over all firms in the group, provide a basis for comparing the efficiency and performance of the parties to the merger with other firms that are similar in terms of size and/or location. These data are indicative, for example, of whether the target firm is relatively inefficient before merger and thus a firm that would seem a good candidate for being improved. Selection of an appropriate control group was left to the discretion of each investigator.

For the post-merger period, the focus of the analysis was on the combined firm relative to a control group. Post-merger data were compared with the pre-merger data to determine what changes occurred in efficiency, performance, and some balance sheet ratios from the pre- to post-merger period. The control group was particularly valuable at this stage because it permits an assessment

¹⁴ It is notable that at least two empirical studies find that efficiency improvements are more likely when both the acquiring and acquired firms are relatively inefficient prior to merger rather than when the acquiring firm is more efficient. See Berger (1997) and De Young (1993).

of whether any observed changes in the combined firm simply reflect changes in the economic environment or instead were unique to the combined firm. The balance sheet ratios were also valuable at this point because they were indicative of whether there had been basic balance sheet changes from the pre- to post-merger period that might be associated with and therefore help explain any observed performance change, rather than the changes being the result of the merger per se.

3.1. Financial ratios

The 16 financial ratios used to examine the efficiency, profitability, and balance sheet structure of the mergers studied were all based on the Consolidated Financial Statements for Bank Holding Companies (Y-9 Report).¹⁵ The standard ratios are based on foreign and domestic data for the consolidated holding company. Consolidated data were used because even though in most cases the vast majority (over 95%) of a firm's assets are in a bank, there are instances where an appreciable amount of a firm's assets are not in a bank. Furthermore, presumably any operations of banking organizations that are centralized in the BHC, rather than the bank, could improve efficiency. By analyzing data for the consolidated organization, any such improvement should be captured.

The analysis of expenses was based on a ratio of expenses to assets or operating revenue rather than absolute expenses in order not to confuse pure expense (cost) reductions with efficiency gains.¹⁶ That is, an absolute expense reduction would not indicate an efficiency gain if assets were reduced proportionately; it would simply reflect a shrinking firm. Examining expenses relative to assets or revenue, however, would control for shrinkage and would be a better indicator of efficiency changes. The denominator in many of the ratios is *average* quarterly assets for the year under review. The rationale for using a simple average of assets over four quarters rather than year-end assets is that expenses (income) represent a flow generated throughout the whole year and the average of the stock of assets at several points during the year is more closely related to these flow measures. Thus, using the value of the stock of assets existing at year-end, which may be considerably larger than assets at the beginning of the year, would not be as accurate a denominator as the average of the assets held throughout the year.

¹⁵ Of course, many other ratios or variants of these 16 ratios could have been used. It was necessary, however, to keep the ratios to a manageable number and select ratios (perhaps from among alternatives) that would serve as reasonable metrics of the subject under consideration.

¹⁶ According to one study, interviews with senior managers at over 50 major banks in various countries indicated that some cost-related *ratio*, even though imperfect, is necessary in order to assess efficiency performance (see Salomon Brothers, 1993).

The ratio of noninterest operating expenses to assets accounts for all of the operating expenses, such as personnel, back-office operations, and branches, that should be directly affected by the cost savings that are frequently cited as resulting from bank mergers, especially horizontal mergers. Consequently, this ratio is of special interest.

Total expenses are arguably the most important because they reflect interest expenses as well as noninterest expenses. Although interest expenses will indeed move in response to general market rates, a bank's interest expenses may be significantly affected by the way it chooses to obtain deposits.¹⁷ For example, a firm may choose to shift from using core deposits (predominantly retail deposits) as a source of funds to using purchased money. Obtaining core deposits tends to incur relatively high noninterest expenses (for retail offices and personnel) and relatively low interest expenses (for retail deposit accounts) while the opposite is true for obtaining purchased money. The expense trade-off made by a bank will be reflected in the total expense figure.

Total expenses to total revenue and noninterest expenses to adjusted operating revenue (net interest income + noninterest income) were analyzed as an alternative to the expenses-to-assets ratios.¹⁸ Using revenue as the denominator provides an alternative view of the expense ratio, reflecting the ability of the firm to generate revenue from its expenditures.¹⁹ One advantage of a revenue-based ratio is that revenues reflect interest rate changes (assets do not) just as total expenses do when used in the numerator. Furthermore, for many banks, revenues reflect income earned off the balance sheet. Such off-balance sheet activity results in expenses but no assets.²⁰ Thus, an expense-to-asset ratio could be misleadingly high for those banks with significant off-balance sheet activities. In contrast, the advantage of using assets as the denominator in the expense ratios is that assets reflect the earnings base of the bank and they are typically not highly variable from one year to the next, whereas revenues tend to be more variable.

¹⁷ Because the expense ratios of the merging firms are analyzed relative to a peer group, this should control for the effects of general interest rate movements on expenses.

¹⁸ Adjusted operating revenue is defined here as total interest income minus total interest expense plus noninterest income.

¹⁹ The ratio of total operating costs to total revenue is the measure almost unanimously preferred by the senior managers at over 50 large banks, according to Salomon Brothers (1993).

²⁰ Among the largest banks, derivatives are especially important off-balance sheet items that may be much larger (as measured by notional value) than total assets. For many other banks, unused commitments such as credit card (personal and business) and home equity lines of credit represent major off-balance sheet items that are sometimes larger in value than assets. Standby and commercial letters of credit represent an important although much smaller source of off-balance sheet items for mostly larger banks. Obviously, many of these activities will not generate the same dollar volume of expenses (or revenues) that an equal amount of assets would generate, but they do generate expenses (and revenues) and the total expense/total revenue ratio will account for this.

The personnel, occupancy, and other noninterest expenses-to-assets ratios are the components of noninterest expenses.²¹ These were analyzed to determine whether the various elements of noninterest expenses change differently from the pre- to post-merger period.

Two profitability or, more precisely, rate-of-return measures were analyzed. One is the ratio of net income (after taxes) to average assets.²² This is an indicator of profitability and a good overall indicator of a banking organization's performance.²³ This ratio illustrates the ability of a firm to generate profits from the assets at its disposal. It is, however, biased upward for some firms due to profits generated from off-balance sheet operations. The net income-to-equity ratio is used as an alternative measure of profitability and is designed to reflect the return to owners' investment. While capturing the return to owners is an attractive feature of this measure, a disadvantage is that the denominator (equity) may vary substantially across firms, even those of identical size, due to discretionary choices by management as to the mix between equity and debt capital as well as the total amount of capital held by a firm.

The five balance sheet ratios are, as noted above, used to determine whether major balance sheet items change from before to after merger and might be responsible for expenses or other performance changes unrelated to efficiency changes. For example, changes in the ratios of total capital to total assets and nonperforming assets to total assets may be indicative of basic changes in risk-taking behavior that may affect performance.²⁴ Increases (decreases) in the ratios of total loans to total assets and C and I loans to total assets would likely result in increases (decreases) in the expense ratios because of the high cost of establishing and maintaining a loan portfolio relative to a portfolio of government securities, with no meaningful implications for operating efficiency. Finally, an increase (decrease) in core deposits to total deposits would

²¹ "Other" noninterest expenses include (a) amortization expense of intangible assets, (b) tax expense based on the bank's gross revenue, and (c) credits resulting from capitalization of imputed interest from internal financing of construction.

²² Net income is defined as net operating income minus applicable income taxes minus extraordinary items and other adjustments.

²³ The Wall Street Journal has referred to return on bank assets as "The key measure of operating efficiency,..." (King, 1993). The importance of return on assets for analyzing bank performance was also noted in Salomon Brothers, *op. cit.*, p. 3, as follows: "... while we view the efficiency measurements as very important to analyzing bank stocks, it takes a back seat to return-on-assets [sic], capital management and credit quality".

²⁴ Total capital is defined as equity capital + subordinated notes and debentures. Nonperforming assets are defined as nonaccruing loans and lease finance receivables. Other real estate owned is not included.

likely result in an increase (decrease) in noninterest expenses and a decrease in interest expenses but would not have implications for operating efficiency.²⁵

Finally, the ratio of off-balance sheet items to total assets is included to provide an indication of an organization's involvement in such activities before and after merger.²⁶ With respect to the derivative components of off-balance sheet activities, because the interest here is in the extent of derivative activities rather than the amount of market value or credit risk (exposure), the notional value rather than gross fair value (which nets out positions held and could be zero even though the firm is extensively involved in derivatives) is used. For a similar reason, the "All Other Off-Balance Sheet" assets and liability items were not netted out.²⁷

3.2. *Econometric analysis*

As an alternative way to analyze the efficiency effects of the mergers, each case study included results of an econometric analysis of three efficiency measures proposed by Berger and Humphrey.²⁸ The three measures – total efficiency, scale efficiency, and X-efficiency – are based on an econometric translog cost function estimate.²⁹ All measures were specified as a percentile ranking of the firm under review relative to all banking firms with more than \$1 billion in assets whether or not they undertook a merger.³⁰ Consequently, all of these measures have a value between zero and one, with one indicating the highest efficiency, which is assigned to the best-practice bank in the sample, and zero indicating lowest performance, which is assigned to the worst performer in

²⁵ Core deposits are defined as demand deposits + NOW accounts + ATS + money market deposit accounts + other savings deposits + time deposits less than \$100,000. NOW accounts and ATS accounts are, like demand deposits, types of transactions accounts but they also earn interest. NOW and ATS accounts are negotiable orders of withdrawal and automatic transfer services.

²⁶ What absolute levels or changes in the ratio of off-balance sheet items to assets may mean for performance in terms of profitability and efficiency is not clear because such activity may simply reflect hedging or it may reflect more speculative activity.

²⁷ The rationale for using notional value in this context may be found in International Swaps and Derivatives Association (1994).

²⁸ The estimation procedure and its application are described in detail in Berger and Humphrey (1992).

²⁹ Scale efficiency measures efficiency solely associated with size, while X-efficiency measures closeness to the efficiency frontier for that size. Total efficiency is the most complete measure and is the product of X-efficiency and scale efficiency.

³⁰ Note that the peer group for the Berger–Humphrey estimates includes all firms with over \$1 billion in assets, whereas the peer groups selected for the ratio analysis of the individual studies include only firms that are more or less similar to the merging firms in terms of their size and/or location.

the sample.³¹ Rankings of firms both before and after merger were computed relative to the peer group in order to assess performance changes. Pre- and post-merger efficiency results using the Berger–Humphrey estimates are based on averaging the ranks over the pre- and post-merger years rather than averaging direct estimates of efficiency over the relevant years.

3.3. *Other information*

Stock price changes of the merging firms around the announcement date, and numerous other sources (in addition to the Call Reports and Bank Holding Company Reports (Y-9)) were also used in the case studies. These other sources included the merger application filed with the federal bank regulator, company annual reports, national and major local newspapers, general business periodicals, banking industry periodicals, industry (and financial) newsletters and reports, and post-merger interviews with officials at the merged banking firm. These information sources are regarded as particularly important in helping to explain whatever findings emerge from the individual studies. That is, these are the sources that should be useful in making informed judgments as to *why* efficiency gains were or were not achieved.

Personal interviews were conducted with officials at eight of the nine post-merger firms. Interviews were conducted only after the analyst had completed a pre-merger analysis, and at least an initial post-merger analysis based on available information sources. The officials interviewed were typically involved in the merger planning or process. In each case, the interviewee(s) was (were) provided with a list of questions in advance of the interview, which usually lasted from 1 to 2 hours. Some of the questions asked were similar for all of the firms, but there were questions unique to individual firms because of special circumstances. The questions generally asked included the following: Are actual cost savings consistent with predicted savings in various operations areas? How do you measure efficiency? What were the biggest challenges or problems you encountered? Did you encounter any surprises (good or bad) following the merger?

4. **Summary of findings**

This section presents a summary of the findings of the nine case studies. It is notable that even among these nine mergers that were selected for study

³¹ The efficiency measures for the pro forma combined firm during the pre-merger period are calculated as weighted averages of the efficiency measures for each of the two firms based on their total assets.

Table 1
Summary of merger case studies: Change in performance

Merger	Acquiring firm more efficient than target	Pre- to post-merger change in performance relative to peers ^a					Stock price change ^b		
		Return on assets	Total expenses		Noninterest expenses		Econometric estimate of "total" efficiency ^c	Acquiring	Acquired
			+ Total assets	+ Total rev.	+ Total assets	+ Adj. oper. rev.			
A	Not clear	Imp	Imp	Imp	Imp	Imp	Imp	Wk	Imp
B	Yes	Imp	Wk	Imp	Imp	Imp	Imp	Wk	Imp
C	Yes	Imp	Wk	Imp	Wk	Imp	Imp	Wk	NA
D	Yes	Imp	Imp	Imp	Wk	Imp	Imp	Imp	NA
E	Yes	Imp	Imp	NC	Imp	Imp	Imp	Wk	Wk
F	Not clear	Imp	Imp	Imp	Imp	Imp	NC	Imp	Imp
G	Not clear	Wk	NC	NC	Imp	Imp	Imp	Imp	Imp
H	Yes	Imp	Wk	Imp	Wk	Imp	NA	Wk	Imp
I	Yes	Wk	Imp	Wk	Imp	Wk	Imp	Imp	Imp

Imp – Improved; Wk – Weakened; NC – No change; NA – Not available.

^a Based on averages of the three years before and after merger. Note that results for all mergers in this table are based on three-year averages, but individual studies may emphasize individual years in assessing performance results. Note that the signs indicate only the direction of change. They do not indicate whether (1) the change was minor or substantial or (2) the resulting *level* of profitability or efficiency is better or worse than the peers (e.g., a merged firm may have improved relative to the peers, but it may still be less efficient or profitable than them).

^b Based on cumulative abnormal return from 10 days before to 10 days after the announcement.

^c This econometric estimate, which is distinct from the expense ratio measures, is described in the text.

because they possessed attributes believed likely to yield efficiency improvements, there was considerable variation in the performance results. Results ranged from no improvement in efficiency to substantial improvement.

A summary of some key information and performance results for each of the nine mergers appears in Table 1. It is important to recognize that the summary table shows only the direction of efficiency and profit results. Firms with the same direction of results may vary substantially in terms of the magnitude of their performance changes relative to peers.³²

4.1. *Main findings*

The reported findings are related to efficiency results (that is, costs relative to assets or revenues), not cost cutting in isolation. The reason, as noted earlier, is that assets and revenues may fall in proportion to costs so that there is no gain in efficiency despite the decline in costs.

The key findings are: (1) all of the studies find that significant cost cutting objectives were achieved or surpassed fairly quickly; (2) four of the nine mergers showed clear efficiency gains relative to peers; and (3) seven of the nine mergers exhibited an improvement in return on assets relative to peers. In addition, the net wealth effect, based on the stock price reaction to the merger announcement, was positive for five of the seven mergers for which data were available.

All of the studies found that the combined firm achieved its cost cutting objectives in a timely fashion. Generally, the largest volume of cost reductions was associated with staff reductions and data processing systems and operations. The reduction in staff costs often accounted for over 50% of the total cost reduction, and in at least one case, reduction in staff costs accounted for nearly two-thirds of the total. In all cases, the savings achieved were of the order of 30–40% of the noninterest expenses of the target. All of the merged firms indicated that the actual savings met or exceeded their expectations. Most of the firms projected that the cost savings would be fully achieved within three years after the merger, with the majority of savings being achieved after two years. All of the acquiring firms had formal plans in place at the time of the merger to meet these deadlines, which they all, reportedly, did. There was considerable variation in the timing of the cost cuts. For example, one merger (E) accomplished nearly all of its cost cuts within one year after the merger whereas

³² Note that for this summary, performance results for all mergers are based on averages of the relevant ratios over all of the pre-merger and post-merger years, respectively. This does not take into account possible trends in performance either before or after the merger, which may be relevant for some mergers. Furthermore, in some instances, a particular performance measure is only very marginally different between the pre- and post-merger periods but that is not captured in the summary.

another merger (F) stretched the cost saving program out over several years. Ultimately, however, both mergers were relatively successful at least in terms of efficiency.

At this point, a brief overview of the findings in Table 1 is presented, followed by a possible explanation of the findings. Even though all of the mergers achieved cost cutting goals, only four of the mergers yielded unambiguous efficiency gains relative to their respective peer groups. This conclusion holds even when focusing only upon the two total expense ratios – total expenses/total assets and total expenses/total revenue.³³ According to the total expenses-to-assets ratio, five of the nine mergers resulted in an improvement in efficiency relative to peers.³⁴ Of the four mergers that did not show improvement based on the assets ratio, three (B, C, and H) decreased in efficiency relative to peers, and one (G) essentially exhibited no change relative to peers. Of these, however, the B, C, and H mergers all showed gains in efficiency relative to peers, based on the total expenses-to-revenue measure, while the G merger again showed essentially no change relative to peers. The I merger was the only one to exhibit a decrease in efficiency relative to peers based on the total expenses-to-revenue ratio, although it showed some gain based on the assets ratio. Finally, all of the mergers, except F, showed an improvement in efficiency relative to peers, based on the econometric estimate of total efficiency.

Most of the mergers were associated with an improvement in return on assets relative to the respective peer groups. Such improvement may have been particularly feasible, because all of the merged firms except G, on average, had a lower return on assets than peers prior to merging. The two mergers that recorded declines in return on assets relative to peers were I and G, although the declines in returns were rather small. It is interesting and perhaps not just coincidental that these were two of the five mergers that resulted in an efficiency loss (or at least no gain) relative to peers based on one of the total expense ratios.

Finally, the stock market tended to have a positive view of the merger announcements. Specifically, the stock price of the *acquiring* firm increased around the announcement date, relative to the market, in five of the mergers (it decreased in four mergers); the stock price of the *target* increased in every merger with available data; and the net wealth effect, based on the change in the aggregate market value of the acquiring and target banks combined, increased in five and declined in two of the mergers for which data were available.³⁵ These results contrast with many studies that have found that the

³³ Efficiency results based on noninterest expenses to total assets and noninterest expenses to adjusted operating revenue were similar.

³⁴ Comparisons in this summary are based on simple averages of the ratios for the three years before and after merger for the appropriate firm(s).

³⁵ It is notable that a recent paper found no relationship between combined abnormal returns and subsequent performance changes (Pilloff, 1996).

stock market generally has a negative reaction to the acquiring firm's stock upon announcement of a merger, including in-market mergers.³⁶ This relatively good stock price performance may reflect the fact that the mergers selected for study were, by design, the type believed most likely to yield efficiency gains. Thus, for at least that reason, they should have been attractive to the market.³⁷

4.2. *Explanation of findings*

Based on the information in Table 1, four out of these nine mergers were clearly successful in improving efficiency, and seven improved profitability relative to peers. Even the two mergers that probably had the weakest overall performance results (I and G) essentially saw only a small decline in performance relative to peers. The data might provide some clues as to the reasons for the successes that were achieved.

In the majority of cases (six out of nine), the acquiring firm was more efficient than its peer group (based on expenses to assets). In contrast, the target firm in the majority of cases (six out of nine) was less efficient than the peer group. The not so surprising result is that in six of the mergers, the acquiring firm was at least marginally more efficient than the target, prior to merger, based on both the expenses-to-asset and expenses-to-revenue ratios. For the other three mergers, the acquiring firm was more efficient based on one of the two measures. In short, we generally had more efficient firms acquiring less efficient firms. This is, however, not necessarily a predictor that a merger will increase efficiency. Thus, in two of the more successful mergers (A and F), the acquirer was *not* unambiguously more efficient than the target. In contrast, the acquirer in I was more efficient than the target, but their merger was not particularly successful in terms of efficiency.

All of the mergers included in this study involved considerable branch overlap; that was one of the criteria for selecting these mergers for analysis. Nevertheless, the degree of office overlap (percentage of offices located in same zip code areas) varied greatly from a low of less than 20% to much higher levels. Both the G and I mergers had relatively weak performance results even though they had the greatest office overlap. It is apparent that the degree of office overlap is not perfectly related to the extent of efficiency gains. It is also clear that

³⁶ Such negative reactions have been typical as indicated by findings from large cross-section "event" studies noted earlier and in investigations of individual mergers as well. An investigation that focused on a fairly large number of individual mergers and found a hostile reaction by the market to many of the deals is SNL Securities (1993).

³⁷ However, the stock price reaction is not necessarily a good indicator of the operating performance results. For example, while the stock price of *both* the acquirer and target in merger G increased, the efficiency and profitability results of this merger were weaker than most of the mergers studied here.

office overlap does not, by itself, guarantee success in increasing efficiency, although it may contribute toward that end.³⁸

The absolute size of the acquiring firm and its size relative to the target also do not seem to be particularly important factors in determining the performance results of the merger. For example, the F merger had relatively good performance results, and it involved two large firms that constituted a merger of equals. The I merger was less successful, as measured here, and it also involved two large firms not too disparate in size. While merger partners of more or less the same size sometimes did relatively well (for example, F) sometimes partners of very different sizes also did relatively well (for example, A).

Various other factors were identified in the case studies as having had a positive influence on a merger, but most of these tended to be case-specific. These factors included a solidly improving economy, good chemistry between CEOs, external pressure for success, and a new headquarters office.

None of the merging banks reported major problems that seriously interfered with achieving their objectives. However, some adverse surprises were encountered. The most frequent and serious problem was unexpected difficulty in integrating data processing systems and operations.

Based on these nine case studies, it does not appear possible to specify with any generality which factor(s) is (are) most likely to cause bank mergers to result in efficiency gains. Although a strong commitment to cutting costs and having acquiring firms that are more efficient than (or as efficient as) the target may be important, they apparently are not sufficient to ensure efficiency gains. For example, all of the acquiring firms studied apparently were committed to and actually did cut costs, yet they did not all show efficiency and profitability gains (for example, I and G). Similarly, with respect to the efficiency of acquirers relative to targets, generalizations are risky.³⁹ As noted above, the acquirer in merger F was not more efficient than the target (more efficient using expenses to assets but less efficient using expenses to revenue), but merger F had relatively strong efficiency results. In contrast, the acquirer in merger I was generally more efficient than the target before the merger, but the performance results of the merger were relatively weak. Thus, a strong commitment to cut-

³⁸ Despite the popularity of this view, it may be suspect. The study by Salomon Brothers concludes that the expectations by analysts for office closure following a merger “has proven unrealistic in competitive markets.” Branch clients may change to a competitor if a branch closure would require they travel more than a small distance (Salomon Brothers, 1993). The potential loss of customers resulting from closing offices is also noted in Zack (1996), and Rhoades (1996). The potential loss of customers from office closure is indicative of the continuing importance of convenient local offices to many retail customers.

³⁹ Interestingly, the bank consultants and analysts we talked with prior to undertaking the studies generally indicated that a relatively efficient acquiring firm was a prerequisite for a merger to result in efficiency improvements.

ting costs and a relatively efficient acquirer may increase the likelihood that a merger will enhance efficiency, but they are clearly not a guarantee. Finally, *both* acquirer and target in merger A were less efficient than their peers prior to their merger, but this was a relatively successful merger in terms of efficiency and profitability gains.

The difficulty of identifying the reasons for the success of a merger is not confined to these case studies. For example, one study that focused on the earnings results of bank mergers noted that “... the success or failure of a given acquisition depends on an interrelated web of factors that extend beyond empirical data.” And further, “... no single factor or group of factors emerged as the cause for their apparent success” (SNL Securities, 1993).

In conclusion, it should be noted that among these nine mergers, all were selected because of basic attributes believed likely to yield efficiency gains; all of the acquirers were committed to cost cutting; and all of the acquirers were more efficient (by at least one measure) than their targets. However, not all of these mergers unambiguously yielded efficiency and profitability improvements despite the favorable characteristics and the significant reduction of noninterest costs achieved in all of these mergers. Consequently, it is not especially surprising that studies using large cross-sections of bank mergers rather than mergers selected for their favorable attributes generally have not found efficiency gains from bank mergers. It is perhaps reasonable to say that a strong commitment to cost cutting (or efficient operation) and a relatively efficient acquiring firm are probably important contributors to a merger having a favorable effect on efficiency. This, of course, avoids saying that either or both of these factors are necessary or sufficient conditions for success. Finally, it may be that there are a variety of benefits that a bank may achieve from a merger even if it does not achieve operating efficiency gains. Such benefits to the bank might include a more diversified deposit and loan base, a different strategic orientation, and a good vehicle for growth.

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